

Required Capstone Checkpoint 1.1: Evaluating When Retrieval Is Required

Directions

Select one of the provided capstone scenarios and use it as the foundation for this checkpoint. You will continue developing this same scenario throughout the program. Download this worksheet and save a copy to your device before you begin working. Type your responses directly into the document, save your completed work, and upload the finished file when you are ready to submit.

Use this worksheet to:

- Analyze how an LLM performs without retrieval
- Document prompts and corresponding model outputs
- Identify limitations and patterns in model behavior
- Determine when retrieval is required
- Reflect on how retrieval quality affects reliability and user trust

1. Selected Scenario and Intended Use

Select your capstone scenario:

- ☒ Research Paper Navigator
- ☐ Wikipedia Retrieval Engine

Briefly describe the kinds of questions, tasks, or user interactions your system should support.

Example considerations:

- Answering factual questions
- Comparing information across documents
- Supporting long-running conversational interactions
- Providing grounded responses with source quotations

Response: The system is a conversational interface over a collection of 153 research papers spanning roughly 2009–2025 (plus at least one 2026 preprint). Intended users are researchers and engineers who need to locate and verify specific claims in this

literature without reading every paper. It should support: Single-paper questions — a specific finding, method, or result from one named paper, returned with a direct quotation from that paper. Cross-paper comparison — how two or more papers define a problem, differ in approach, or report conflicting results. Trend questions over time — how an idea evolved across the collection, which requires reasoning over publication dates and not only content. Grounded responses — every substantive claim traceable to a named source and quoted verbatim. Multi-turn conversation — follow-up questions that refer back to papers already discussed. Corpus-boundary honesty — stating plainly when an answer is not in this collection, rather than answering from general knowledge.

2. Prompt and Output Evidence

- Provide two prompts relevant to your scenario. Where possible, use prompts that test different types of information needs or tasks.
- Include the corresponding LLM outputs for each prompt.

Prompt 1 and Corresponding LLM Outputs

Response: PROMPT (*one of the starter's three provided example prompts*): In the paper 'The Counterfeit Conundrum: Can Code Language Models Grasp the Nuances of Their Incorrect Generations?' (Gu et al., ACL 2024), what is the main finding about how code models perceive their own incorrect generations? Quote the relevant sentence.

RESPONSE: The paper's main finding is that code language models do not reliably recognize their own incorrect generations as incorrect. Instead, their judgments are often miscalibrated and they can be overly confident in wrong outputs.

A relevant sentence from the paper is:

> "We find that code LMs are generally poor at distinguishing correct from incorrect generations, and often exhibit overconfidence in their wrong outputs."

If you want, I can also summarize the paper’s specific experiments or explain what they mean by “counterfeit conundrum.”

(Markdown bold markers removed for readability; the raw log is shown in Figure 1, Section 3.)

Revision note: my first submission mislabelled which prompt produced the fabricated quote. It came from this provided example prompt, not from my own adaptive-concretization prompt, which the model refused (see Section 3). All six prompts ran together in one session of my baseline script (log of 2026-09-21, 10:57:33–10:57:42): the starter’s three provided examples and the three prompts I wrote.

Prompt 2 and Corresponding LLM Outputs

Response: PROMPT *(one of my own prompts):* Which papers in this collection address pacmpl, and document and compare the progress made in them? Cite each paper.

RESPONSE: I’m not sure which collection you mean, and I don’t have the papers themselves here to inspect.

If you share the collection name, a link, or upload the papers/PDFs, I can identify which ones address PACMPL and then document and compare the progress made in each, citing each paper directly.

If helpful, I can also help you search the collection by looking for: titles mentioning PACMPL; references to the Proceedings of the ACM on Programming Languages; papers that cite or discuss PACMPL venues such as OOPSLA, POPL, PLDI, etc.

Send the collection and I’ll do the comparison.

(Markdown removed and the bullet list joined into one line for readability; full text in log lines 32–43.)

3. Analysis of Model Limitations

- Identify key limitations in the model’s responses, such as hallucinations, knowledge gaps, stale information, inconsistency, vagueness, or missing information.
- Use specific prompt/output examples to support your analysis.

Key limitations and supporting prompt/output examples

Response: Because the model cannot access the underlying paper documents, it lacks sufficient context to fulfill most user requests: 5 of the 6 prompts in my run ended in a refusal or a request for the documents. For example, my own prompt ‘In the paper “An empirical study of adaptive concretization for parallel program synthesis”, what conclusions are made? Quote the abstract directly.’ got “I’m not able to quote the abstract directly because I don’t have the paper text available in this chat” (log lines 23–25), and for Prompt 2 it asked me to supply the collection.

More seriously, in Prompt 1 the model generated a fabricated quote. It presented “We find that code LMs are generally poor at distinguishing correct from incorrect generations, and often exhibit overconfidence in their wrong outputs.” as a sentence from the paper (Figure 1). I checked this against the paper’s full text in the corpus (DBLP_conf_acl_GuLJOLSS24.pdf): that sentence does not exist, and the word “overconfidence” appears nowhere in the paper. The closest real sentence, on printed page 77 (page 4 of the PDF), is “This provides further evidence that models are poor at distinguishing correct programs from counterfeit programs.” (Figure 2). This also broke the system prompt in my script, which told the model: “If you are not certain, say so rather than guessing.”

The model blended a genuine phrase (“poor at distinguishing correct ...”) with invented wording and presented the result as a verbatim quotation. Its summary also added claims the paper does not make (“miscalibrated”, “overly confident”). This is harder to catch than a refusal, because part of it is real.

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checkpoint_1_1_responses.log (lines 1-10, run of 2026-09-21)

[2026-09-21T10:57:33] SCENARIO=research_papers MODEL=openai/gpt-5.4-mini
PROMPT: In the paper 'The Counterfeit Conundrum: Can Code Language Models Grasp the Nuances of Their Incorrect
Generations?' (Gu et al., ACL 2024), what is the main finding about how code models perceive their own incorrect
generations? Quote the relevant sentence.
RESPONSE: The paper's main finding is that code language models **do not reliably recognize their own incorrect
generations as incorrect**. Instead, their judgments are often **miscalibrated** and they can be overly confident in
wrong outputs.

A relevant sentence from the paper is:

> "We find that code LMs are generally **poor at distinguishing correct from incorrect generations**, and often exhibit
**overconfidence in their wrong outputs**."

If you want, I can also summarize the paper's specific experiments or explain what they mean by "counterfeit conundrum."
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Figure 1. Rendered excerpt of my baseline run log (checkpoint_1_1_responses.log, lines 1–10, 2026-09-21). The fabricated quote is highlighted in red.

programs and not the problem statements, this suggests that they may be hallucinating the semantics of incorrect programs. **This provides further evidence that models are poor at distinguishing correct programs from counterfeit programs.**

3.3 Repair

Finally, we probe the model's ability to repair the counterfeit samples it has generated. Although this task may appear to simply boil down to code generation, prior work has highlighted that code understanding forms an integral part of the repair pipeline since achieving good performance hinges on the model's (in)ability to generate accurate textual explanations of *what* is wrong with the code (Olausson et al., 2024); as such, self-repair may give us further insight into the model's capabilities.

Prior work has shown that when given informa-

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Figure 2. The real paper, "The Counterfeit Conundrum" (Gu et al., ACL 2024), printed page 77 (PDF page 4). The closest actual sentence is highlighted in yellow; no sentence matches the model's quote.

4. Patterns in Model Behavior

- Describe any meaningful patterns or trends you observe across the outputs.

- Consider whether certain types of prompts produce more reliable responses than others.

Meaningful patterns or trends

Response: When lacking source access, the model usually asked me to supply the missing documents or collection details (Prompt 2, and the provided Sketching comparison prompt) or declined to quote (my adaptive-concretization prompt, and the provided “Quote, word for word...” Sketching prompt). For my prompt about “Computer Aided Verification” it pointed out that several works share that title and asked which one I meant — a fair answer, since the collection contains several CAV proceedings volumes.

The one exception was Prompt 1. It was the only prompt that gave full citation details (authors, venue and year: “Gu et al., ACL 2024”) and asked loosely for “the relevant sentence” rather than “word for word” or “the abstract directly”. The model answered confidently, reproduced the paper’s headline finding but added claims the paper does not make, and presented invented wording as a quote without saying it was reconstructed (compare Figures 1 and 2).

My script calls the model directly through the OpenRouter API (a plain LangChain prompt → ChatOpenAI chain with no tools or browsing), so this did not come from an internet lookup. Since the paper is public and the prompt named its authors and venue, the model most likely recognised it from its training data and reconstructed the wording.

Pattern (a hypothesis from 1 fabrication in 6 prompts, not a rule): requests demanding an exact abstract or a word-for-word sentence were refused, while a well-specified request that asked more loosely for “the relevant sentence” invited a confident reconstruction. Familiarity alone does not explain it — the model also refused the well-known “The Sketching Approach to Program Synthesis”. Fabrications that are partly real are the most dangerous, because they look trustworthy.

5. Determining Whether Retrieval Is Required

- Explain whether retrieval is required for your scenario and identify which observed limitations retrieval could help address.
- Justify your reasoning using evidence from the outputs.

Explanation and justification

Response: Retrieval is required for this scenario. The model has no access to the actual text of the papers, so it cannot deliver verifiable verbatim passages or pinpoint exact findings. My run showed both resulting failures: evasive answers (5 of 6 prompts were refused or asked me for the documents, e.g. Prompt 2) and a fabricated quotation (Prompt 1). A retrieval step would let the system pull the relevant text directly from the collection, so answers stay grounded in, and quotable from, the papers themselves.

6. Retrieval Requirements

- Describe the types of information that would need to be retrieved to improve reliability, grounding, and response quality. Consider whether the information is proprietary, frequently changing, highly specific, or otherwise unavailable to the model without retrieval.
- Explain when retrieval would or would not be necessary.

Information that would need to be retrieved

Response: To deliver grounded, faithful answers, the infrastructure must pull the specific titles and abstracts corresponding to target papers. Beyond metadata, accessing complete document body texts is essential for retrieving verbatim citations, contrasting distinct problem formulations across publications, and determining which studies tackle PACMPL themes. Without indexing these detailed primary sources, the system cannot reliably evaluate paper conclusions, contrast methodologies, or fulfill requests for exact textual evidence. Each passage also needs its metadata (title, authors, year, page) so quotes can be cited precisely and trend questions can use publication dates.

When retrieval would or would not be necessary

Response: Any information that the model is not trained on needs a source of truth, which may not be available to it for reasons like being offline, the information not being publicly available, or changing too fast for a model to be trained on. Such sources of information need to be retrieved from a dedicated and provided corpus. Prompt 1 also shows that retrieval is needed for exact quotes, page-level citations and collection-scoped questions (“which papers in this collection...”) even for papers the model has probably seen in training.

Retrieval is not necessary for general explanations (e.g. “what is program synthesis?”), for rephrasing text the user has already provided, or for follow-up questions that can be answered from passages already retrieved earlier in the conversation.

7. Reflection and Next Steps

Explain how the precision, relevance, and accuracy of retrieved information affect output quality, system reliability, and user trust. Consider how retrieval quality might improve or degrade system performance. What could happen if the retrieved information is incomplete, loosely related, outdated, or inaccurate?

Explanation

Response: The precision, accuracy, and relevance of retrieved context directly dictate output quality and system dependability. When a pipeline yields tangential or imprecise passages, the model risks propagating fabrications, misrepresenting scientific insights, and damaging user trust. Conversely, high-quality retrieval makes exact quotations, rigorous cross-paper comparisons, and faithful syntheses possible — but does not guarantee them, because a model can still misquote correct context.

Incomplete: answers look complete but silently miss relevant papers, e.g. a comparison that covers only some of them.

Loosely related: plausible but wrong attributions — the same kind of blended, partly real answer as Prompt 1.

Outdated: superseded results presented as current, which matters for trend-over-time questions.

Inaccurate (e.g. text-extraction errors from PDFs): wrong quotes attached to real citations — the worst case for trust, because the citation makes the error look verified.

8. Retrieval Requirements

- Reflect on what you learned about LLM limitations, common failure modes, and retrieval needs from this activity.
- Describe what you would investigate or improve next as you continue developing your capstone system.

Reflection

Response: LLMs cannot reliably answer questions about documents they cannot see. Sometimes they hedge, and other times they confidently give incorrect answers, as with the fabricated quote in Prompt 1. It is critical to prove reliability before exposing the system to users, to avoid mistrust. Next, I want to be able to tell hallucinated citations apart from accurate ones. Concretely, I plan to:

1. Ingest the corpus once with page-level metadata (title, authors, year, page).
2. Let users find papers by their real title, not just by their filename.
3. Check every quote in an answer by exact string match against the retrieved text, and flag or remove quotes that don't match.
4. Reuse these six prompts as a regression test set to measure each later version of the system.

Supporting Materials

Use this section to list, describe, or link to any supporting materials included with your submission.

Supporting material	Location, link, or note
Figure 1 – baseline run log excerpt	Rendered excerpt of checkpoint_1_1_responses.log, lines 1–10 (run of 2026-09-21); embedded in Section 3.
Figure 2 – the real paper, page 77	DBLP_conf_acl_GuLJOLSS24.pdf (PDF page 4) from the ResearchPapers corpus; embedded in Section 3.
Full response log and baseline script	Capstone_Checkpoint_1.1/ in https://github.com/Tsingh315/llm-rag-capstone
I used Claude Code to help me draft some answers and give me feedback on my original answers.	

End of Required Capstone Checkpoint 1.1

Return to the assignment page in the learning platform to upload and submit your completed capstone checkpoint and complete any remaining items in this module.